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Subject: Linear Algebra
Class Tests
Date: August 1, 2026

Class Test 1: $\boxed{28/07/2026}$
Write down the general solution of the following matrix given in REF form: $A\vec{x} = \vec{b}$

$$\left[\begin{array}{ccccc|c} 2 & 0 & 0 & 5 & 2 & 4 \\ 0 & 0 & 1 & 3 & 7 & 6 \\ 0 & 0 & 0 & -1 & 1 & 8 \end{array}\right]$$

Also find two solutions of $A\vec{x} = \vec{0}$ where they are not scalar multiples of each other.

Proof. The matrix is already in REF form. So we simply solve from bottom and utimately get the general solution as follows:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 22 - 3.5\alpha \\ \beta \\ 30 - 10\alpha \\ \alpha - 8 \\ \alpha \end{bmatrix} = \alpha \begin{bmatrix} -3.5 \\ 0 \\ -10 \\ 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 22 \\ 0 \\ 30 \\ -8 \\ 0 \end{bmatrix}, \alpha, \beta \in F$$

$\square$

For second part, simply consider these two

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

or something else idc.

Class Test 2: hewwo!! $> \omega <$

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